

Conforms to Regulation (EC) No. 1907/2006 (REACH), Annex II, as amended by Regulation (EU) No. 453/2010

SAFETY DATA SHEET

FOR INDUSTRIAL USE ONLY

BAKELITE® PF8431SP SACK 20KG

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : BAKELITE® PF8431SP SACK 20KG
SDS Number : 1614300
Product type : Novolak Resin
Other means of identification : BAKELITE HARZ 8431 SP

1.2 Relevant identified uses of the substance or mixture and uses advised against

Product use : Abrasive Applications

1.3 Details of the supplier of the safety data sheet

Manufacturer/Supplier/Importer : Hexion GmbH
Gennaer Str. 2-4
58642 Iserlohn Olona
Nrth Rhine Westfalia, Germany

Contact person : 4information@hexion.com

Telephone : General information
+31 6 52 511079

1.4

Emergency telephone number
Supplier : CARECHEM24
Telephone number : +44 (0) 1235 239 670

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Skin Sens. 1 H317

Classification according to Directive 1999/45/EC [DPD]

Classification : R43

Physical/chemical hazards	:	Not applicable.
Human health hazards	:	May cause sensitization by skin contact.
Environmental hazards	:	Not applicable.

See Section 16 for the full text of the R phrases or H statements declared above.

2.2 Label elements

Hazard pictograms	:	
Signal word	:	Warning
Hazard statements	:	May cause an allergic skin reaction.

Precautionary statements

Prevention	:	Wear protective gloves.
Response	:	IF ON SKIN: Wash with plenty of soap and water. If skin irritation or rash occurs: Get medical attention.
Storage	:	Not applicable.
Disposal	:	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Hazardous ingredients	:	methenamine
Supplemental label elements	:	Not applicable.

2.3 Other hazards

Substance meets the criteria for PBT according to Regulation (EC) No. 1907/2006, Annex XIII	:	Not applicable.
Substance meets the criteria for vPvB according to Regulation (EC) No. 1907/2006, Annex XIII	:	Not applicable.
Other hazards which do not result in classification	:	Unclassified Hazard - Combustible Dust Combustible dust when finely divided and suspended in air. Fine dust clouds may form explosive mixtures with air. Product can explode if dust cloud is formed and ignited. Minimize airborne dust. Eliminate all fire/ignition sources including static discharges near product/package. Prevent dust accumulation. Refer to Handling Section 7 of the MSDS for more information. Handling and/or processing of this material may generate a dust which can cause mechanical irritation of the eyes, skin, nose and throat.

SECTION 3: Composition/information on ingredients

Substance/mixture : Mixture

Product/ingredient name	Identifiers	% by weight	<u>Classification</u>		Type
			67/548/EEC	Regulation (EC) No. 1272/2008 [CLP]	
methenamine	RRN : 01-2119474895-20 EC:202-905-8 CAS : 100-97-0 Index:612-101-00-2	>=10 - <15	F; R11 R43	Flam. Sol. 2, H228 Skin Sens. 1, H317	[1]
Phenol	RRN : 01-2119471329-32 EC:203-632-7 CAS : 108-95-2 Index:604-001-00-2	>=0.1 - <1	Muta.Cat.3; R68 T; R23/24/25 C; R34 Xn; R48/20/21/22	Acute Tox. 3, H301 Acute Tox. 3, H311 Acute Tox. 3, H331 Skin Corr./Irrit. 1B, H314 Eye Dam./Irrit. 1, H318 Muta. 2, H341 STOT RE 2, H373	[1][2]

Type

[1] Substance classified with a health or environmental hazard

[2] Substance with a workplace exposure limit

[3] Substance meets the criteria for PBT according to Regulation (EC) No. 1907/2006, Annex XIII

[4] Substance meets the criteria for vPvB according to Regulation (EC) No. 1907/2006, Annex XIII

See Section 16 for the full text of the R phrases or H statements declared above.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

SECTION 4: First aid measures

4.1 Description of first aid measures

- Eye contact** : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Continue to rinse for at least 10 minutes. Get medical attention if irritation occurs.
- Inhalation** : Remove victim to fresh air and keep at rest in a position comfortable for breathing. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Get medical attention if adverse health effects persist or are severe. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.

- Skin contact** : Wash with plenty of soap and water. Remove contaminated clothing and shoes. Wash contaminated clothing thoroughly with water before removing it, or wear gloves. Continue to rinse for at least 10 minutes. Get medical attention. In the event of any complaints or symptoms, avoid further exposure. Wash clothing before reuse. Clean shoes thoroughly before reuse.
- Ingestion** : Wash out mouth with water. Remove dentures if any. Remove victim to fresh air and keep at rest in a position comfortable for breathing. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Get medical attention if adverse health effects persist or are severe. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.
- Protection of first aid personnel** : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

- Eye contact** : Exposure to airborne concentrations above statutory or recommended exposure limits may cause irritation of the eyes.
- Inhalation** : Exposure to airborne concentrations above statutory or recommended exposure limits may cause irritation of the nose, throat and lungs. Exposure to decomposition products may cause a health hazard. Serious effects may be delayed following exposure.
- Skin contact** : May cause an allergic skin reaction.
- Ingestion** : No known significant effects or critical hazards.

Over-exposure signs/symptoms

- Eye contact** : Adverse symptoms may include the following:
irritation
redness
- Inhalation** : Adverse symptoms may include the following:
respiratory tract irritation
coughing
- Skin contact** : Adverse symptoms may include the following:
irritation
redness
- Ingestion** : No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

- Notes to physician** : In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.
- Specific treatments** : No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

- Suitable extinguishing media** : Use water spray or mist, dry chemical, foam or CO₂.
Unsuitable extinguishing media : Do not use water jet.

5.2 Special hazards arising from the substance or mixture

- Hazards from the substance or mixture** : Combustible solid that burns. Fine dust clouds may form explosive mixtures with air.
Hazardous thermal decomposition products : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
nitrogen oxides

5.3 Advice for firefighters

- Special protective actions for fire-fighters** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Move containers from fire area if this can be done without risk. Use water spray to keep fire-exposed containers cool.
- Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.
- Additional information** : Organic powders when finely divided over a range of concentrations regardless of particulate size or shape and suspended in air or some other oxidizing medium may form explosive dust-air mixtures and result in a fire or dust explosion (including secondary explosions). The ATEX Directive defines combustible powders as less than 500 microns in diameter. When processed with flammable liquids/vapors/mists, ignitable (hybrid) mixtures may be formed with combustible dusts. Ignitable mixtures will increase the rate of explosion pressure rise and the MIE will be lower than the pure dust in air mixture. The Lower Explosive Limit (LEL) of the vapor/dust mixture will be lower than the individual LELs for the vapors/mists or dusts. See NFPA 77 for additional guidance.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

- For non-emergency personnel** : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Minimize airborne dust and eliminate all fire/ignition sources. Clean up spill as soon as possible using procedures described below. Avoid breathing dust. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment.
- For emergency responders** : If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

- 6.2 Environmental precautions** : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.3 Methods and material for containment and cleaning up

- Small spill** : Move containers from spill area. Do not use air hoses for cleaning. Minimize dry sweeping to avoid generation of dust clouds. Vacuum dust-accumulating surfaces and remove to a chemical disposal area. Use spark-proof tools and explosion-proof equipment. Vacuums with explosion-proof motors should be used. Dispose of via a licensed waste disposal contractor.
- Large spill** : Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Avoid creating dusty conditions and prevent wind dispersal. Do not use air hoses for cleaning. Minimize dry sweeping to avoid generation of dust clouds. Vacuum dust-accumulating surfaces and remove to a chemical disposal area. Use spark-proof tools and explosion-proof equipment. Vacuums with explosion-proof motors should be used. Dispose of via a licensed waste disposal contractor. Note: see section 1 of SDS for emergency contact information and section 13 of SDS for waste disposal.

- 6.4 Reference to other sections** : See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

- Protective measures** : Put on appropriate personal protective equipment (see section 8 of SDS). Persons with a history of skin sensitization problems should not be employed in any process in which this product is used. Do not get in eyes or on skin or clothing. Do not ingest. Avoid breathing dust. Avoid the creation of dust when handling and avoid all possible sources of ignition (spark or flame). Prevent dust accumulation. Use only with adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Electrical equipment and lighting should be protected to appropriate standards to prevent dust coming into contact with hot surfaces, sparks or other ignition sources. Take precautionary measures against electrostatic discharges. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before transferring material. Empty containers retain product residue and can be hazardous. Do not reuse container.

COMBUSTIBLE DUST HANDLING PROCEDURES:

Combustible dusts at sufficient concentrations can form explosive mixtures with air. High dust concentrations should be avoided. Follow US NFPA Standard 654, "Standard for the Prevention of Fire

and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids," UK HSE Guidance HSG 103, approved Codes of Practice (ACOPS) established for Explosive Atmospheres under the ATEX Directive 1999/92/EC for worker protection and ATEX Directive 94/9/EC that regulates equipment and protection systems used in potentially explosive atmospheres or other national guidance on safe handling of combustible dusts. Train workers in the recognition and prevention of hazards associated with combustible dust in the plant.

Minimize airborne dust and eliminate all ignition sources. Keep away from heat, hot surfaces, sparks, and flame. Establish good housekeeping practices. Remove dust accumulations on a regular basis by vacuuming or gentle sweeping to avoid creating dust clouds. Use continuous suction at points of dust generation to capture and minimize the accumulation of dusts. Particular attention should be given to overhead and hidden horizontal surfaces to minimize the probability of a "secondary" explosion. According to NFPA Standard 654, dust layers 1/32 in.(0.8 mm) thick can be sufficient to warrant immediate cleaning of the area.

Control sources of static electricity. This product or the package itself can accumulate static charges, and static discharge can be a source of ignition. Solids handling systems must be designed in accordance with applicable NFPA standards (including 654 and 77) and other national guidance. Do not empty directly into flammable solvents or in the presence of flammable vapors. The operator, the packaging container and all equipment must be grounded with electrical bonding and grounding systems. Plastic bags and plastics cannot be grounded, and antistatic bags do not completely protect against development of static charges.

Advice on general occupational hygiene : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a segregated and approved area. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see section 10 of SDS) and food and drink. Eliminate all ignition sources. Separate from oxidizing materials. Keep away from heat, hot surfaces, sparks and flame. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination.

7.3 Specific end use(s)

Recommendations : Not available
Industrial sector specific solutions : Not available

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational exposure limits

No exposure limit value known.

Recommended monitoring procedures : If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

DNELs/DMELs

Product/ingredient name	Type	Exposure	Value	Population	Effects
Phenol	DNEL	Long term Inhalation	8.0 mg/m ³	Workers	Systemic
Phenol	DNEL	Long term Dermal	0.4 mg/kg bw/day	General	Systemic
Phenol	DNEL	Long term Oral	0.4 mg/kg bw/day	General	Systemic
Phenol	DNEL	Long term Inhalation	1.32 mg/m ³	General	Systemic

DNEL/DMEL Summary : Not available

PNECs

Product/ingredient name	Type	Compartment Detail	Value	Method Detail
Phenol	PNEC	Fresh water	7.7 µg/l	
Phenol	PNEC	Marine	0.77 µg/l	
Phenol	PNEC	Sewage Treatment Plant	2.1 mg/l	
Phenol	PNEC	Fresh water sediment	91.5 µg/kg dw	
Phenol	PNEC	Marine water sediment	9.15 µg/kg dw	
Phenol	PNEC	Soil	136 µg/kg dw	

PNEC Summary : Not available

Derived No-Effect Levels' (DNEL's) and Predicted No-Effect Concentrations' (PNEC's)

Explanatory note:

REACH requires manufacturers and importers to establish and report 'Derived No-Effect Levels' (DNEL's) for humans by inhalation, ingestion and dermal routes of exposure and 'Predicted No-Effect Concentrations' (PNEC's) for environmental exposure. DNEL's and PNEC's are established by the registrant without an official consultation process, and are not intended to be directly used for setting workplace or general population exposure limits. They are primarily used as input values in running Quantitative Risk Assessment models (like the ECETOC-TRA model).

Due to differences in calculation methodology the DNEL will tend to be lower (sometimes significantly) than any corresponding health-based OEL for that chemical substance. Further although DNEL's (and PNEC's) are an indication for setting risk reduction measures, it should be recognized that these limits do not have the same regulatory application as officially endorsed governmental OEL's.

8.2 Exposure controls

Appropriate engineering controls : Use only with adequate ventilation. If user operations generate dust,

fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. The engineering controls also need to keep gas, vapor or dust concentrations below any lower explosive limits. Use explosion-proof ventilation equipment.

Individual protection measures

- Hygiene measures** : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.
- Eye/face protection** : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: safety glasses with side-shields. If operating conditions cause high dust concentrations to be produced, use dust goggles.

Skin protection

- Hand protection** : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers. In the case of mixtures, consisting of several substances, the protection time of the gloves cannot be accurately estimated.
- Body protection** : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product., For PPE selection see National Fire Protection Association (NFPA) 2113, Standard on Selection, Care, Use and Maintenance of Flame-Resistant Garments for Protection of Industrial Personnel Against Flash Fire.
- Other skin protection** : Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
- Respiratory protection** : Use a properly fitted, air-purifying or air-fed respirator complying with an approved standard if a risk assessment indicates this is necessary. Respirator selection must be based on known or anticipated exposure levels, the hazards of the product and the safe working limits of the selected respirator. Where air-filtering respirators are suitable, select an appropriate combination of mask and filter, ABEK-P3 (EN14387)
- Environmental exposure controls** : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance

Physical state	:	Powder
Color	:	Yellowish.
Odor	:	characteristic.
Odor threshold	:	Not available
pH	:	6.5 - 7.5 [Conc. (% w/w): 200 g/l] @ 20 °C (ISO 8975)
Melting point/freezing point	:	70 - 110 °C (ISO 3146 A)
Initial boiling point and boiling range	:	Not available
Flash point	:	Not defined for solids
Evaporation rate	:	Not available
Flammability (solid, gas)	:	Not available
Burning time	:	Not available
Burning rate	:	Not available
Upper/lower flammability or explosive limits	:	Lower: Not available Upper: Not available
Vapor pressure	:	Not available
Vapor density	:	Not available
Relative density	:	Not available
Bulk density	:	300 - 500 kg/m ³ (DIN 53466)
Solubility(ies)	:	Not available
Solubility in water	:	Partial
Partition coefficient: n-octanol/water	:	Not available
Auto-ignition temperature	:	Not available
Decomposition temperature	:	Not available
Viscosity	:	Dynamic: Not available Kinematic: Not available
Explosive properties	:	Not available
Oxidizing properties	:	Not available

9.2 Other information

No additional information.

SECTION 10: Stability and reactivity

10.1 Reactivity	:	Stable under normal conditions.
10.2 Chemical stability	:	The product is stable.
10.3 Possibility of hazardous reactions	:	Under normal conditions of storage and use, hazardous reactions will not occur.
10.4 Conditions to avoid	:	Avoid the creation of dust when handling and avoid all possible sources of ignition (spark or flame). Take precautionary measures against electrostatic discharges. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding

containers and equipment before transferring material. Prevent dust accumulation. See Section 7 Handling.

10.5 Incompatible materials : Reactive or incompatible with the following materials:
acids
oxidizing materials

10.6 Hazardous decomposition products : Under normal conditions of storage and use, hazardous decomposition products should not be produced.

During processing, vapors of phenol, ammonia and formaldehyde may be released.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
methenamine				
	LD50 Oral	Rat	> 20,000 mg/kg	-
	LD50 Dermal	Rat	> 2,000 mg/kg	-
Phenol				
	LD50 Oral	Rat	317 mg/kg	-
Remarks - Oral:	In studies conducted in a manner similar to current O.E.C.D. test guideline, the rat LD50 ranged 340 - 650 mg/kg of body wt.			
	LC50 Inhalation	Rat	0.316 mg/l	-
	LD50 Dermal	Rabbit	630 mg/kg	-
Remarks - Dermal:	In studies conducted in a manner similar to the current O.E.C.D. test guideline, the rat dermal LD50 ranged 525 - 707 mg/kg of body wt and the rabbit LD50 was 850 mg/kg of body wt.			

Conclusion/Summary : Not available

Acute toxicity estimates

Not available

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure	Observation
Phenol	Skin - -	Rat	> 4		-
	eyes - Cornea opacity	Rabbit	> 3		-

Conclusion/Summary

Skin : Not available
eyes : Not available
Respiratory : Not available

Sensitization

Product/ingredient name	Route of exposure	Species	Result
Phenol	Skin	-	-
Remarks:	Not sensitizing in an O.E.C.D. test guideline no. 406 guinea pig Buehler study.		

	However, the Challenge dose was only 1%. Not sensitizing in a human Maximization test conducted with an induction dose of 2% and a Challenge dose of 1%.
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Conclusion/Summary

Skin : Not available
Respiratory : Not available

Mutagenicity

Product/ingredient name	Test	Experiment	Result
Phenol	-	; -	-
Remarks:	Not a bacterial mutagen in O.E.C.D. test guideline no. 471 Ames/Salmonella mutation assays conducted up to cytotoxic dose levels with and without S9 metabolic activation. Positive for induction of micronuclei (chromosome damage) in Chinese hamster ovary (CHO) cells at 3-7-fold the control background frequency when tested to cytotoxic dose levels. Positive for the induction of chromosome aberrations in CHO cells only with S9 metabolic activation. Induced a 2-3-fold increase of the gene-mutation frequency in independent studies in mouse lymphoma cells with and without S9 metabolic activation. Evidence for the induction of sister-chromatid-exchanges (SCEs), DNA strand breaks and DNA adducts also reported. In independent mouse bone marrow micronucleus studies, weakly (statistically) positive by I.P. injection, but not by oral gavage at myelotoxic doses of approximately 300 mg/kg/day. Research suggest that the mechanism of micronucleus formation may involve hypothermia a near lethal doses. No DNA adducts detected in rat bone marrow or liver following 4 doses of 75 mg/kg/day. Currently classified as Mutagen Category 2.		

Conclusion/Summary : Not available

Carcinogenicity

Product/ingredient name	Result	Species	Dose	Exposure
Phenol	- - - -	-		
Remarks:	In long-term oral drinking water cancer bioassays (NIH/NCI) in rats and mice no evidence of carcinogenicity in mice and female rats. The increased tumor incidence observed in male rats was considered not treatment related. No evidence of tumors in wild-type and transgenic TG.AC mice following 20 weeks of treatment (2 days/week). In mice treated twice/week with 25 ul 20% phenol (corrosive) for 32 weeks 7/18 developed skin papillomas. Limited evidence for tumor promoting activity on mouse skin at corrosive 20% concentrations.			

Conclusion/Summary : Not available

Reproductive toxicity

Product/ingredient name	Maternal toxicity	Fertility	Development toxin	Species	Dose	Exposure
Phenol	-	-	-	-	-	-
Remarks:	In an O.E.C.D. test guideline no. 416 rat two-generation oral drinking water study the adult and reproductive NOAEL was approximately 70 mg/kg/day. There was a significant group mean reduction of body weight, feed consumption and water consumption at the high concentration of 5000 ppm (~ 300 mg/kg/day) in both generations. Group mean pup body weight and survival were significantly reduced at 5000 ppm. These adverse findings are believed to be to the drinking water palitability of the high dose phenol.					

Conclusion/Summary : Not available

Teratogenicity

Product/ingredient name	Result	Species	Dose	Exposure
Phenol	- - -	-	-	-

Remarks:	O.E.C.D. test guideline no. 414 developmental toxicity studies were conducted in the rat and mouse by oral gavage. The NOAEL for both maternal and developmental toxicity in the mouse was 140 mg/kg/day. There were maternal mortalities and a significant reduction in mean maternal body weight at 280 mg/kg/day. Also, clinical signs including tremors and ataxia were observed at 280 mg/kg/day. Mean fetal body weight was significantly reduced at the high dose of 280 mg/kg/day. In the rat the maternal NOAEL was 60 mg/kg/day due to significantly reduced mean body weight at 120 and 360 mg/kg/day. The developmental effects NOAEL was 120 mg/kg/day due to a significant reduction in mean fetal body weight and ossification sites at the high dose of 360 mg/kg/day. These data suggest a significant role for maternal toxicity in the adverse developmental effects observed.
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Conclusion/Summary : Not available

Specific target organ toxicity (single exposure)

Not available

Specific target organ toxicity (repeated exposure)

Not available

Aspiration hazard

Not available

Information on the likely routes of exposure : Not available

Potential acute health effects

Eye contact : Exposure to airborne concentrations above statutory or recommended exposure limits may cause irritation of the eyes.
Inhalation : No known significant effects or critical hazards.
Skin contact : May cause an allergic skin reaction.
Ingestion : Exposure to airborne concentrations above statutory or recommended exposure limits may cause irritation of the nose, throat and lungs. Exposure to decomposition products may cause a health hazard. Serious effects may be delayed following exposure.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact : Adverse symptoms may include the following:
irritation
redness
Inhalation : Adverse symptoms may include the following:
respiratory tract irritation
coughing
Skin contact : Adverse symptoms may include the following:
irritation
redness
Ingestion : No specific data.

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects : Not available
Potential delayed effects : Not available

Long term exposure

Potential immediate effects : Not available
Potential delayed effects : Not available

Potential chronic health effects

Conclusion/Summary : Not available

General : Repeated or prolonged inhalation of dust may lead to chronic respiratory irritation. Once sensitized, a severe allergic reaction may occur when subsequently exposed to very low levels.

Carcinogenicity : No known significant effects or critical hazards.
Mutagenicity : No known significant effects or critical hazards.
Teratogenicity : No known significant effects or critical hazards.
Developmental effects : No known significant effects or critical hazards.
Fertility effects : No known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Product/ingredient name	Result	Species	Exposure
methenamine			
	Acute LC50 49,800,000 µg/l Fresh water	Fish - Fathead minnow	96 h
	Acute EC50 36,000,000 µg/l Fresh water	Aquatic invertebrates. Water flea	48 h
Phenol			
	Acute LC50 8.9 mg/l Fresh water	Fish - Rainbow trout,donaldson trout	96 h
	Acute NOEC 0.077 mg/l Fresh water	Fish - Carp	60 d
	Acute EC50 3.1 mg/l Fresh water	Aquatic invertebrates. Water flea	48 h
	Acute NOEC 0.16 mg/l Fresh water	Aquatic invertebrates. Water flea	16 d
	Acute EC50 61.1 mg/l Fresh water	Aquatic plants - Microalgae	96 h
	Acute EC50 21 mg/l Fresh water	Micro-organism - Soil organisms	24 h
	Chronic NOEC 2.2 mg/l Fresh water	Aquatic invertebrates. Water flea	2 d

Conclusion/Summary : Not available

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
Phenol		-		
Remarks:	In two independent O.E.C.D. test guideline no. 301C Modified MITI studies the level of biodegradation was 62% within 4.2 days and 85% after 14 days of contact. In an O.E.C.D. test guideline no. 302B study the level of biodegradation was 100% after 6 days.			

Conclusion/Summary : Not available

12.3 Bioaccumulative potential

Not available

12.3 Bioaccumulative potential

Product/ingredient name	LogPow	BCF	Potential
methenamine	-2.18	-	low
Phenol	1.5	17.5	low

12.4 Mobility in soil

- Soil/water partition coefficient (KOC)** : Not available
- Mobility** : Not available

12.5 Results of PBT and vPvB assessment

- PBT** : P: Not available
B: Not available
T: Not available
- vPvB** : vP: Not available
vB: Not available

- 12.6 Other adverse effects** : No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product

- Methods of disposal** : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction.
- Hazardous waste** : The classification of the product may meet the criteria for a hazardous waste.

Packaging

- Methods of disposal** : The generation of waste should be avoided or minimized wherever possible. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible.
- Special precautions** : This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Regulatory information ADR/ADN	14.1. UN number	14.2. UN proper shipping name	14.3. Transport hazard class(es)	14.4. Packing group
		Not dangerous goods		
RID		Not dangerous goods		
ICAO/IATA		Not dangerous goods		
IMO/IMDG		Not dangerous goods		

14.5. Environmental hazards

Environmentally hazardous and/or Marine Pollutant : No.

14.6 Special precautions for user : Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.'

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1907/2006 (REACH)
Annex XIV - List of substances subject to authorization
Substances of very high concern

Carcinogen: Not listed
Mutagen: Not listed
Toxic to reproduction: Not listed
PBT: Not listed
vPvB: Not listed

Other EU regulations

REACH Status : The substance(s) in this product has (have) been Pre-Registered and/or Registered, or are exempted from registration, according to Regulation (EC) No. 1907/2006 (REACH).

Aerosol dispensers : Not applicable.
Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles : Not applicable.

EU - Prior Informed Consent. List of chemicals subject to the international PIC procedure (Annex I - Part 1) : Not listed

EU - Prior Informed Consent. List of chemicals subject to the international PIC procedure (Annex I - Part 2) : Not listed

EU - Prior Informed Consent. : Not listed
List of chemicals subject to the international PIC procedure (Annex I - Part 3)

AOX : Not available

Product/ingredient name	Carcinogenic effects	Mutagenic effects	Developmental effects	Fertility effects
Phenol		Muta.Cat.3; R68 Muta. 2, H341		

Seveso II Directive

This product is controlled under the Seveso II Directive.

National regulations

Hazardous incident ordinance : Not applicable.
Hazard class for water : WGK 1, Appendix No. 4
Technical instruction on air quality control : Number 5.2.1:
Number 5.2.5:
Number 5.2.5:

International regulations

International lists : Australia inventory (AICS) All components are listed or exempted.
Canada inventory All components are listed or exempted.
Japan inventory All components are listed or exempted.
China inventory (IECSC) All components are listed or exempted.
Korea inventory All components are listed or exempted.
New Zealand Inventory (NZIoC) All components are listed or exempted.
Philippines inventory (PICCS) All components are listed or exempted.
United States inventory (TSCA 8b) All components are listed or exempted.

Chemical Weapons Convention : Not listed
List Schedule I Chemicals
Chemical Weapons Convention : Not listed
List Schedule II Chemicals
Chemical Weapons Convention : Not listed
List Schedule III Chemicals

15.2 Chemical Safety Assessment : This product contains substances for which Chemical Safety Assessments are still required.

SECTION 16: Other information

Abbreviations and acronyms : ATE = Acute Toxicity Estimate
CLP = Classification, Labelling and Packaging Regulation
[Regulation (EC) No. 1272/2008]
DNEL = Derived No Effect Level
DMEL = Derived Minimal Effect Level
EUH statement = CLP-specific Hazard statement
PNEC = Predicted No Effect Concentration
RRN = REACH Registration Number
PBT = Persistent, Bioaccumulative and Toxic
vPvB = Very Persistent and Very Bioaccumulative

Procedure used to derive the classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Classification	Justification
Skin Sens. 1, H317	Calculation method

Full text of abbreviated H statements	:	H228	Flammable solid.
		H317	May cause an allergic skin reaction.

Full text of classifications [CLP/GHS]	:	Flam. Sol. 2, H228	FLAMMABLE SOLIDS - Category 2
		Skin Sens. 1, H317	SKIN SENSITIZATION - Category 1

Full text of abbreviated R phrases : R11- Highly flammable.
R43- May cause sensitization by skin contact.

Full text of classifications [DSD/DPD] : F - Highly flammable

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